PAPER_617: UQFF SCm Laurent Series 26D Expansion

Unified Quantum Field Framework (UQFF)

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Abstract

The superconducting medium field SCm is expressed as a degree-26 Laurent series: $SCm = \lambda \cdot UA \cdot (1 - 1/t) + \sum_{m=0}^{26} b_m t^{-m}$. The negative-power expansion in $1/t^m$ encodes the complete time-reversal asymmetry of the superconducting phase across 26 cosmic epochs. Early-universe divergence is bounded by the 26! factorial threshold; late-time behavior converges to $\lambda UA + b_0$.

§1. Introduction

The scalar superconducting medium field SCm governs the coupling amplitudes in all UQFF components. Its time-dependence encodes the phase transition history of the universe across 26 epochs. A complete Laurent series representation replaces the previous approximation $SCm \approx \lambda UA(1 - 1/t)$.

§2. Laurent Series Formulation

$$\boxed{SCm = \lambda \cdot UA \cdot \left(1 - \frac{1}{t}\right) + \sum_{m=0}^{26} \frac{b_m}{t^m}}$$

2.1 Base Term

$$SCm_{\text{base}} = \lambda \cdot UA \cdot \left(1 - \frac{1}{t}\right)$$

Vanishes at $t=1$ (present cosmic epoch). Asymptotes to $\lambda \cdot UA$ as $t \to \infty$.

2.2 Laurent Series Terms

$$\sum_{m=0}^{26} b_m t^{-m} = b_0 + \frac{b_1}{t} + \frac{b_2}{t^2} + \cdots + \frac{b_{26}}{t^{26}}$$

The 26th derivative of the m-th term:

$$\frac{d^{26}}{dt^{26}}\!\left(b_m t^{-m}\right) = \frac{(m+25)!}{(m-1)!} \cdot \frac{b_m}{t^{m+26}}$$

For $m=26$: $\frac{51!}{25!} \cdot b_{26}/t^{52}$

2.3 Asymptotic Behavior

t value	SCm behavior
$t = 1$	$SCm \approx \sum b_m$ (present-day sum)
$t \to 0^+$	$SCm \to \infty$ (big-bang divergence, bounded by $26!$)
$t \to \infty$	$SCm \to \lambda UA + b_0$ (late-universe asymptote)

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§3. VDS Coefficient Assignment

The coefficients b_m are assigned from the Vacuum Density Series (VDS) digit expansion: $b_m = \pi_{\text{digit}(m)} \times 10^{-m}$, ensuring:

- Non-repeating values (irrational π digits)
- Monotonically decreasing amplitudes
- Laurent convergence radius > 1 (physical time domain)

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§4. VDS / DVP / BH26 Connections

- **VDS**: b_m coefficients are π-indexed vacuum density series weights per cosmic epoch.
- **DVP**: Laurent convergence radius equals the DVP prime gap bound for the series.
- **BH26**: The 26th term b_{26}/t^{26} corresponds to BH26 epoch-26 temporal separation.

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§5. Conclusions

The degree-26 Laurent expansion of SCm captures the full superconducting phase history from the Big Bang to the present epoch. The factorial 26! threshold ensures that early- universe divergence remains physically bounded, and the unique b_m assignment via VDS/π-digits guarantees no two epochs share the same coupling amplitude.

Class: `UQFFSCmLaurentSeries26DExpansionCalculator` (#204, CP4 v5.17) **Source**: `grok_share_79fdf5367d1.txt` (161 lines, March 29, 2026)